# Classical Field Theory — Exercise Sheet 24

*A higher Lax flow and operator domains*

**Solutions**

Use $D_x = \partial_x$ and the ordered commutator $[B, L] = BL - LB$. All operator calculations in the first problem are on $C_c^\infty(\mathbb{R}, \mathbb{C})$, with smooth real coefficient $u$. A displayed differential expression does not specify its Hilbert-space domain.

**Exercise 24.1 (Determining a fifth-order Lax flow).** Let $L = -D_x^2 + u$. For real constants $r_1, \dots, r_6$ consider

$$B_5 = 16D_x^5 + r_1 u D_x^3 + r_2 u' D_x^2 + (r_3 u'' + r_4 u^2) D_x + r_5 u''' + r_6 u u'.$$

Products with $u$ and its derivatives here mean multiplication before the indicated derivative of the test.

(a) Determine the constants so that $[B_5, L]$ is multiplication for every smooth $u$. Find the resulting scalar coefficient and write the corresponding evolution equation $L_t = [B_5, L]$.

(b) Prove that the resulting $B_5$ is formally skew-symmetric on compact smooth tests. Distinguish this from a statement about a closed skew-adjoint operator on $L^2$.

**Solution.** For a test $\psi$, write $[B_5, L]\psi = \sum_{j=0}^{4} C_j \psi^{(j)}$. Leibniz's rule gives the coefficients of differentiated tests:

$$\begin{aligned}
C_4 &= (80 + 2r_1)u', \\
C_3 &= (160 + r_1 + 2r_2)u'', \\
C_2 &= (160 + r_2 + 2r_3)u''' + (3r_1 + 4r_4)uu', \\
C_1 &= (80 + r_3 + 2r_5)u'''' \\
&\quad + (2r_2 + 2r_4 + 2r_6)(u')^2 + (3r_1 + 2r_4 + 2r_6)uu''.
\end{aligned}$$

For example, $[16D_x^5, u]$ contributes $80u'\psi^{(4)} + 160u''\psi''' + 160u'''\psi'' + 80u''''\psi' + 16u^{(5)}\psi$. Each other term is obtained by the same product rule and commutation with $-D_x^2$ or multiplication by $u$.

To force $C_j = 0$ for every smooth $u$, its finite jets at any point may be prescribed independently by a polynomial times a cutoff. The first two equations give $r_1 = -40$ and $r_2 = -60$. The two terms of $C_2$ then give $r_3 = -50$ and $r_4 = 30$. The $u''''$ and $(u')^2$ terms give $r_5 = -15$ and $r_6 = 30$. The last $uu''$ equation is then also zero. The undifferentiated coefficient before substitution is

$$C_0 = (16 + r_5)u^{(5)} + (r_1 + r_6)uu''' + (r_2 + r_3 + 3r_6)u'u'' + r_4 u^2 u'.$$

Thus the unique operator in this ansatz is

$$B_5 = 16D_x^5 - 40uD_x^3 - 60u'D_x^2 + (30u^2 - 50u'')D_x + 30uu' - 15u''',$$

and its commutator is multiplication by

$$K_5[u] = u^{(5)} - 10uu''' - 20u'u'' + 30u^2 u'.$$

The equation $L_t = [B_5, L]$ is $u_t = K_5[u]$. No existence result for that evolution is asserted by this calculation.

For the formal adjoint, compact integration by parts gives $(aD_x^j)^* = (-D_x)^j \circ a$ for real $a$. Let $c_1 = 30u^2 - 50u''$ and $c_0 = 30uu' - 15u'''$. Expansion gives

$$\begin{aligned} B_5^* &= -16D_x^5 + 40D_x^3 \circ u - 60D_x^2 \circ u' - D_x \circ c_1 + c_0 \\ &= -16D_x^5 + 40uD_x^3 + 60u'D_x^2 - c_1 D_x \\ &\quad + 40u''' - 60u''' - c_1' + c_0. \end{aligned}$$

Since $c_1' = 60uu' - 50u'''$, the last coefficient equals $15u''' - 30uu' = -c_0$. Hence $B_5^* = -B_5$ on tests. This means $(f, B_5 g) = -(B_5 f, g)$ for compact smooth $f, g$. It does not identify a closed operator or its Hilbert adjoint domain.

**Exercise 24.2 (Constancy for a sufficiently regular eigenbranch).** Let $u \in C^1(I; \mathcal{S}(\mathbb{R}, \mathbb{R}))$ solve $u_t = K_5[u]$ on an open interval $I$. Suppose $\lambda \in C^1(I; \mathbb{R})$ and $g \in C^1(I; H^7(\mathbb{R}, \mathbb{C}))$, with $g(t) \neq 0$ and $L(t)g(t) = \lambda(t)g(t)$ at every time. The Sobolev space $H^7$ uses the Fourier norm $\|\langle k \rangle^7 \widehat{g}\|_2$.

Use the following operator fact as a *given*: a real bounded multiplication perturbation of $-D_x^2$ is self-adjoint on the common domain $H^2(\mathbb{R})$. This is the relative-bound-zero case of Kato, *Perturbation Theory for Linear Operators*, V, Theorem 4.3, p. 287. Prove that $\lambda$ is constant. Specify the domain needed for moving $L$ across an inner product. Does this calculation require simplicity or isolation, and does it construct a branch or prove a theorem about the whole spectrum?

**Solution.** Multiplication by $u$ and all its spatial derivatives preserves the Sobolev spaces used here, since those coefficient derivatives are bounded. The eigenvalue equation $g'' = (u - \lambda)g$ successively gives $g \in H^m$ for every integer $m$, so it has a smooth representative. Only the assumed $H^7$ regularity is needed for $B_5 g \in H^2$. The equation $Lg = \lambda g$ also implies $B_5(Lg) = \lambda B_5 g$. Both $LB_5 g$ and $B_5 Lg$ belong to $L^2$.

The assumed time regularity permits differentiation into $L^2$:

$$u_t g + Lg_t = \lambda' g + \lambda g_t.$$

The real eigenvalue and self-adjointness give $(g, Lg_t) = (Lg, g_t) = \lambda(g, g_t)$. Subtracting these terms yields

$$\lambda' \|g\|_2^2 = (g, u_t g) = (g, [B_5, L]g) = \lambda(g, B_5 g) - (Lg, B_5 g) = 0.$$

The commutator equality is the finite differential formula proved in the first problem, now with both compositions justified on this function. Moving $L$ uses $g \in H^2$ and $B_5 g \in H^2$. Since $g$ is nonzero at every time, $\lambda' = 0$ throughout $I$.

The proof did not need an eigenspace of dimension one or an isolated eigenvalue. Such properties are important in the separate problem of obtaining differentiable eigenbranches. Here the branch and its strong regularity were hypotheses. Neither their existence nor an evolution operator on all of $L^2$ has been constructed.

**Exercise 24.3 (The same interior expression with moving endpoints).** Fix $\ell > 0$ and a smooth positive nonconstant function $\kappa(t)$. On $L^2(0,\ell)$ consider the expression $L = -\partial_x^2$ with domain

$$\mathcal{D}_t = \left\{ f \in H^2(0,\ell) : f(0) = 0, \quad f'(\ell) = \beta(t) f(\ell) \right\}, \qquad \beta(t) = \kappa(t)\coth(\kappa(t)\ell).$$

The trace values in this formula have their usual continuous Sobolev representatives on a finite interval.

(a) Show that this realization is symmetric by its endpoint Green formula. Find a nonzero smooth eigenfunction with a nonconstant eigenvalue. Self-adjointness is not required for this task.

(b) With $B = 0$, verify $L_t\psi = [B, L]\psi = 0$ on every compactly supported interior test. Explain why this does not contradict the preceding conditional eigenbranch result or establish constancy of the varying eigenvalue.

**Solution.** For $f, g \in \mathcal{D}_t$, integration by parts gives

$$(f, Lg) - (Lf, g) = [\overline{f'}g - \overline{f}g']_0^\ell.$$

At zero both undifferentiated values vanish. At $\ell$, both derivatives equal the same real coefficient $\beta(t)$ times their values, so the two products cancel. This proves symmetry on the stated domain.

The function $g(t,x) = \sinh(\kappa(t)x)$ obeys $g(t,0) = 0$ and

$$g_x(t,\ell) = \kappa\cosh(\kappa\ell) = \kappa\coth(\kappa\ell)\sinh(\kappa\ell) = \beta(t)g(t,\ell).$$

It is nonzero and $-g_{xx} = -\kappa(t)^2 g$. Its eigenvalue is therefore nonconstant when $\kappa$ is nonconstant.

The differential coefficient of $L$ is independent of time, so on every compact interior test $L_t\psi = 0 = [0, L]\psi$. This is only a common interior-test identity. Differentiation of the endpoint condition instead gives

$$(g_t)'(\ell) - \beta(t)g_t(\ell) = \beta'(t)g(\ell).$$

Whenever $\kappa'(t) \neq 0$, its right side is nonzero. Indeed, with $z = \kappa\ell > 0$, $\mathrm{d}(\kappa\coth(\kappa\ell))/\mathrm{d}\kappa = (\sinh z\cosh z - z)/\sinh^2 z > 0$. The numerator is zero at zero and has derivative $2\sinh^2 z > 0$. Thus $g_t$ need not belong to $\mathcal{D}_t$. The cancellation using a fixed self-adjoint domain in the preceding problem is unavailable. A zero identity for the interior differential expression alone does not control time-dependent boundary domains.